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Energy Input-Output Analysis for Onion Production in Major Onion Growing Areas of Karnataka State

A. Carolin Rathinakumari*, G. Senthil Kumaran, S. A. Venu, B. S. Ashok Suraj, Murugesh Huchagoudar, and A. R. Surendra

ICAR-Indian Institute of Horticultural Research, Bengaluru, Karnataka – 560 089; *Corresponding Author's E-mail Address: rathina72.kumari@gmail.com

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ABSTRACT

Energy and agricultural production are interlinked. Crop cultivation process involves energy use as well as energy production. A study was thus carried out to estimate the energy input and output for onion cultivation in Karnataka State of India. Onion is an agricultural produce found in almost every household in India. Farm operations in onion cultivation include land preparation, sowing, manure and fertilizer application, plant protection, irrigation, harvesting, detopping and grading. Data for this study was collected through a structured survey schedule and converted into the respective energy equivalents. Energy parameters in terms of direct energy (included men, women, tractor, diesel, and electricity), indirect energy (included seed, fertilizer, pesticide, and machinery) and output energy were calculated. Fertilizer energy (8,849.26 MJ ha^{-1}) was found to be maximum utilized energy source in onion cultivation followed by electricity (3,983.44 MJ ha^{-1}), diesel (2,992.14 MJ ha^{-1}) and women (2,966.78 MJ ha^{-1}) contributing about 44.98%, 20.25%, 15.21% and 15.08% of input energy, respectively. Fertilizer application (8,968.41 MJ ha^{-1}) had the highest share of energy consumption operation followed by irrigation (4,135.90 MJ ha^{-1}) and land preparation (2,904.98 MJ ha^{-1}). Onion cultivation consumed a total input energy of 19,782.29 MJ ha^{-1} (54.10% direct energy and 45.90% indirect energy). The output energy was calculated as 21,826.40 MJ ha^{-1} with an energy ratio of 1.11. The study elucidated that building a bridge between farmer's awareness as well as education and developed agricultural technologies by research institutions can only be a helpful strategy to increase energy productivity and efficiency of overall agricultural production in the near future.

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1. INTRODUCTION

Onion (*Allium Cepa* L.) is one of the most widely cultivated vegetables in the world. India shares 14% of global onion production and ranks second after China (Nagendra & Ramappa, 2019). Area, production and productivity of onion during 2021- 2022 were reported to be 1914 thousand ha, 31273 thousand MT and 16339 kg ha^{-1}, respectively (GOI, 2022). Major onion growing states in India are Andhra Pradesh, Bihar, Gujarat, Karnataka, Madhya Pradesh, Maharashtra, Rajasthan, Tamil Nadu, Uttar Pradesh and West Bengal. Karnataka is one of the leading states in the cultivation of onion next to Maharashtra and Madhya Pradesh occupying 14% of area and 10% of the total production of onion in the country (GOI, 2022). The area, production and productivity of onion in Karnataka during 2021-2022, were 230.4 thousand ha, 2660.68 thousand MT and 11548 kg ha^{-1}, respectively (GOI, 2022).

Crop cultivation is a process of energy transfer and energy conversion. Agriculture and energy use are thus closely linked (Esengun *et al.,* 2007). Increasing human population, decreasing arable land and technological

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advancements are impetus for ever increasing energy use in agriculture. It depends mostly on availability of energy sources in any particular region. The energy use pattern also varies with the agro-climatic zones. (Swaminathan *et al.,* 2020). Energy requirements in agriculture are divided into two groups – direct and indirect. The direct sources of energy are those that are used directly like man power, bullocks, fuel/electricity use by diesel engines, electric motors, power tillers, and tractors. The indirect sources of energy are those which are not used directly on the farm but are used during manufacture and transportation of fertilizers, chemicals and machinery. Seeds, manures (farmyard and poultry), chemicals, fertilizers, and machinery can be classified under indirect sources of energy (Ozkan *et al.,* 2007; Swaminathan *et al.,* 2020).

Many researchers have studied energy use patterns for different crop production systems. *viz.*, wheat, maize, soybean (Sartori *et al.*, 2005); tomato (Hatirli *et al.*, 2006; Esengun *et al.*, 2007); soybean (Mandal *et al.*, 2002); green gram (Tripathi *et al.*, 2013) cultivations; chickpea in Vertisols of semi–arid tropics (Patil *et al.*, 2014). Ozkan *et al.* (2004) studied about greenhouse vegetable production in Turkey. Energy use efficiency and economic analysis of pulses production in Iran was carried out by Koocheki *et al.* (2011).

Sartori *et al.* (2005) analysed the production cost for wheat, maize and soyabean and the role of European Union subsidies on farm strategies for important emerging management systems namely conservation farming (CF) and organic farming (OF) systems. They reported 50% higher total energy input per unit surface area in the CF system for the 3-yr rotation as compared to that of in OF system. In CF systems the net energy per hectare was 58% greater as compared to the OF systems. They also reported higher average energy conversion index (ECI) and efficiency of energy use (EEU) in the OF system (ECI:23% and EEU:31%) than in the CF system, and particularly in soya bean (ECI of 25%; EEU of 31%) and in wheat (ECI of 53%; EEU of 83%). Esengun *et al. (*2007) estimated that the amount of energy consumed in stake-tomato production was 96,957.36 MJ ha^{-1} with energy input–output ratio of 0.80 and energy productivity of 1.00 kg MJ^{-1}. About 76% of the total energy inputs used in stake-tomato production was non-renewable while only about 22% was renewable. While studying the energy inputs and crop yield relationship in greenhouse tomato production, Hartirli *et al.* (2006) reported that small size farms were more energy efficient than large ones in terms of output–input ratio. In the surveyed farms, output–input ratio, specific energy and energy productivity were 1.2, 12 380.3 MJ t^{-1} and 0.09 kg MJ^{-1}, respectively. Patil *et al.* (2014) reported net energy benefit, energy use efficiency, energy productivity and specific energy for chickpea (Cicer arietinum L.) cultivars in Vertisols of semi-arid tropics as 45,800 MJ ha^{-1}, 10.35, 0.304 kg MJ^{-1} and 3.38 MJ kg^{-1}, respectively. Mandal *et al.* (2002) reported soybean as the most energy-investment intensive crop with highest specific energy for soybean (9173 MJ t^{-1} grain) followed by mustard (8912 MJ t^{-1} seed), chickpea (7190 MJ t^{-1} grain) and wheat (6646 MJ t^{-1} grain). However, energy intensiveness (MJ Rs^{-1}) was higher in wheat (1.40) followed by mustard (1.11), soybean (0.89) and chickpea (0.87) and the soybean–wheat system (1.13) emerged as the most energy-intensive system compared to soybean–mustard (0.97) and soybean–chickpea (0.88). Thus, quantification of energy use and expenditure as well as identification of energy intensive farm management practices can significantly impact crop production. Though many investigations were carried out on energy use pattern in agriculture, there is no systematic study on the energy use pattern of onion cultivation in India.

Karnataka state comprises of ten agro-climatic zones, out of which, onion is cultivated majorly in five agro-climatic zones *viz*., Northern Dry Zone, North Transition Zone, Southern Transition Zone, Central Dry Zone, Southern Dry Zone. There are three types of onions *viz*., rose onion, common onion and aggregatum type onions that are cultivated in Karnataka. Cultivation of each type primarily differs in four main aspects, namely, planting system, planting method, plant geometry, and irrigation method. The yield for each type varies significantly compared to other. Energy use pattern also varies depending on the cultivation practices, type of onion and agro-climatic conditions. Rathinakumari *et al*. (2022) reported energy productivity of aggregatum type onion cultivated in Southern Dry Zone of Karnataka State as 0.54 kg MJ^{-1} and suggested that consumption of fertilizer, diesel and electricity should be optimized so that the energy ratio can further be increased. In Northern Dry Zone, common type of onion is grown majorly and also this zone contributes to major share (66.45%) in total onion production (GoK, 2022). A grasp of energy input–output pattern of common onion cultivation appears essential to enhance the energy ratio as well to identify the gaps in environmentally cleaner onion production system in such region. Hence, present study

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was taken up to analyse energy input–output in onion cultivation in major onion growing areas of Karnataka State, and suggest suitable interventions to increase the energy ratio.

2. MATERIALS AND METHODS

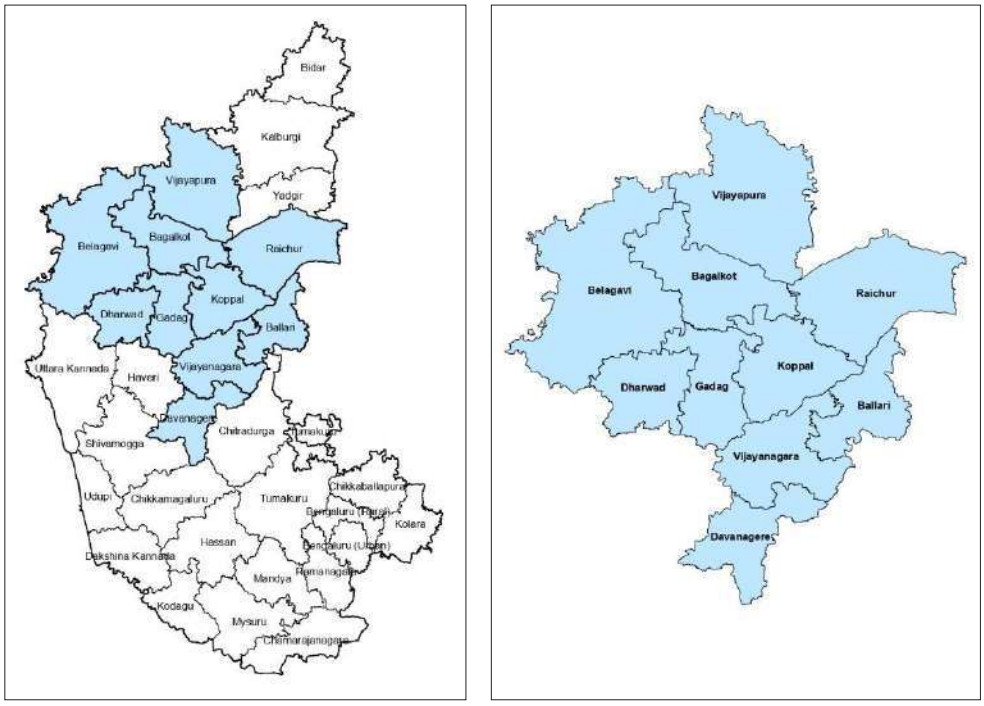
2.1 Study Area

Among the ten agro-climatic zones of Karnataka State, the highest onion producing agro-climatic zone i.e., Northern Dry Zone was selected for the present study. The study area is located in Northern part of Karnataka, which lies between 13°47' and 17°30' N and 74°05' and 77°36' E with an altitude ranging from 75 to 1104 m above the mean sea level (MSL) (Fig. 1). It is a dry zone and semi-arid region, comprising of Vijayapura, Belagavi, Bagalkot, Gadag, Koppal, Bellary, Davanagere, Vijayanagara, Raichur and Dharwad Districts of Karnataka state covering an area of 67,884.59 km^2. The mean annual rainfall of this region ranges from 420 to 3700 mm. The main soil types in this area are shallow to very shallow black soils, medium deep to deep black soils, and red soils with clayey to loamy in texture.

2.2 Data Collection

The data was collected by interactive survey through structured schedules comprising of two parts; i) Schedule – I: Basic information of households in the selected village and ii) Schedule – III: Detailed information of onion cultivation of randomly selected onion growing farmers (N=320). Sixteen villages were randomly selected from Northern Dry Zone and twenty onion growing farmers from each village were randomly selected (Chandra *et al.*, 2021). The villages included in this study are Arakeri (211), Bidiri (177), Halagili (191), Machakanur (205), Sunaga (198), Lalasangi (203), Sathalagaon (188), Bannikoppa (199), Kadadahalli (187), Hebballi (168), Nayakanur (216), Shalawadi (204), Sotakanal (191), Sulla (231), Thondyala (219) and Uttangi (176). The number provided inside the parenthesis are the number of households surveyed for primary data (Schedule –I) in the respective village. Out of the total 3164 households surveyed in different villages, 20 onion growing farmers were randomly selected from each village for detailed survey (Schedule –III).

Various farm operations involved in onion cultivation are land preparation, sowing, irrigation, weeding, chemical spraying, fertilizer application, harvesting, detopping and grading. Required information for cultivation of onion such as use of men, women, tractor, diesel, electricity, for different farm operations (in terms of total working hours) and use of seed, fertilizer, crop protection chemical (insecticides,



(a) Northern Dry Zone of Karnataka State (b) Study area under Northern Dry Zone

Fig. 1. Location map of Northern Dry Zone of Karnataka state

fungicides, herbicides) (in terms of quantity) were recorded (schedule-III). Motor horse power (HP) was recorded for an individual farmer field and converted to kilowatt (kW). Electricity consumption for irrigation was estimated from total time consumed and power used and expressed as kWh. Information on type and kind of agri-machinery, operation time and volume of diesel fuel consumed in each field operation in onion cultivation were collected. Summarized data on the amount, frequency and method of applications of various inputs are presented in Table 1. The output energy is onion yield; hence crop yield data was collected.

2.3 Energy Calculations

2.3.1 Input energy

Input energy from men, women, tractor, diesel, electricity, seed, manure, chemical fertilizers and plant protection chemicals were calculated using the specific energy coefficients. The input energy used in engaging agri-machinery (mobile or stationary) was calculated from the total weight, useful life, energy coefficient and duration of operation of respective machine. The physical unit obtained from various sources for one acre (4000 m^2) for different farm operations were converted into their energy value using respective energy equivalents and expressed as MJ ha^{-1}. The energy requirement for electricity consumed in groundwater pumping for irrigation purpose was calculated using the Horse Power of the submersible pump-set and duration of pump-set operation. Energy equivalents (Table 2) were used to calculate the energy required in each farm operation of onion cultivation.

The energy consumption for all sources *viz*., men, women, tractor, diesel, electricity, seed, fertilizer, pesticide and machinery were determined using the following equations. (Lohan *et al*., 2023 and Ozkan *et al*., 2004).

Table 1. Farm operations for onion production in the study area

Land preparation	
Implements used	Single plough (1); 5 tyne cultivator (2); 9 tyne cultivator (1); Rotavator (1)
Sowing	
Method	Broadcasting
Plant geometry	row-to-row: 15-20 cm plant-to-plant: 10-15 cm.
Crop Variety	Arka Kalyan, Poona Fursungi, Local, Bhima Super, Garva
Seed treatment Manual	Trichoderma viride or Pseudomonas fluorescens to prevent soil-borne diseases.
Season (Kharif and Rabi)	Kharif: June to July Rabi: October to November
Irrigation	
Method of irrigation	Drip, Sprinkler, flood and rainfed irrigation.
Frequency	3-5 days once depending upon soil and weather conditions.
Tools/Equipments/Machinery used	Spade, submersible pump
Crop protection measures	
Weed management	Manual using hand hoe
Weed, pest and decease control	Battery operated knapsack sprayer
Fertilizers application	
Manure application	Manual broadcasting
Fertilizer application	Manual and water soluble fertilizer through drip
Harvesting	Manual using hand hoe
Detopping	Manual using sickle/serrated sickle
Grading	Manual

Note: Number in parenthesis indicated number of passes performed.

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Table 2. Energy equivalents used for onion cultivation in NDZ of Karnataka state

Energy Types	Unit	Energy Equivalent, MJ	Reference
Human Labour			
Men	Man-hour	1.96	Singh and Mittal (1992) Chaichana *et al*. (2014)
Women	Women-hour	1.57	Mandal *et al*. (2002)
Fuel			
Diesel	Litre	56.31	
Chemical Fertilizers			
Nitrogen (N)	kg	60.60	
Phosphate (P_2O_5)	kg	11.10	
Potassium (K_2O)	kg	6.70	
Urea	kg	27.88	
$SuPO_4$	kg	1.77	
MoP	kg	4.02	Singh and Mittal (1992) Chaichana *et al*. (2014)
DAP	kg	15.97	
Chemicals	kg	120	
FYM	kg	0.30	
Insecticide	kg	199	
Fungicides	kg	92	
Herbicides	kg	238	
Electricity	kWh	11.93	
Electric motor	kg	64.65	
Onion			
Seeds (Onions)	kg	14.7	Ozkan *et al*. (2004)
Onion Yield	kg	1.6	

$$M_E = \left(\frac{N_L \times W_H}{U_A}\right) \times EE_M \quad ...(1)$$

$$W_E = \left(\frac{N_L \times W_H}{U_A}\right) \times EE_W \quad ...(2)$$

$$T_E = \left(\frac{W_T}{L_T} \times \frac{W_H}{U_A}\right) \times EE_T \quad ...(3)$$

$$D_E = \left(\frac{F_C}{U_A}\right) \times EE_D \quad ...(4)$$

$$E_E = \left(\frac{E_C \times P_{OD}}{U_A}\right) \times EE_E \quad ...(5)$$

$$S_E = \left(\frac{Q_S}{U_A}\right) \times EE_S \quad ...(6)$$

$$F_E = \left(\frac{Q_F}{U_A}\right) \times EE_F \quad ...(7)$$

$$P_E = \left(\frac{Q_P}{U_A}\right) \times EE_P \quad ...(8)$$

$$MA_E = \left(\frac{W_{MA} \times W_H}{L_{MA} \times U_A}\right) \times EE_{MA} \quad ...(9)$$

where, M_E is men energy input (MJ ha^{-1}), N_L is No. of labours (persons), W_H is working hours (h), U_A is unit area (ha), EE_M is energy equivalent of men (MJ h^{-1}), W_E is women energy input (MJ ha^{-1}), EE_W is energy equivalent of women (MJ h^{-1}), T_E is tractor energy input (MJ ha^{-1}), W_T is weight of tractor (kg), L_T is lifespan of tractor (h), EE_T is energy equivalent of tractor (MJ kg^{-1}), D_E is diesel energy (MJ ha^{-1}), F_C is fuel consumption (l), EE_D is energy equivalent of diesel (MJ l^{-1}), E_E is electricity energy (MJ ha^{-1}), E_C is electricity consumption (kWh h^{-1}), P_{OD} is pump operation duration (h), EE_E is energy equivalent of electricity (MJ kWh^{-1}), S_E is seed energy, Q_S is quantity of seeds (kg), EE_S is energy equivalent of seed (MJ kg^{-1}), F_E is fertilizer energy (MJ ha^{-1}), Q_F is quantity of fertilizer (kg), EE_F is energy equivalent of fertilizer (MJ kg^{-1}), P_E is pesticide energy input (MJ ha^{-1}), Q_P is quantity of pesticide (kg), EE_P is energy equivalent of pesticide (MJ kg^{-1}), MA_E is machinery energy (MJ

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ha^{-1}), W_{MA} is weight of specific machine (kg), L_{MA} is life of machine (h), EE_{MA} is energy equivalent of machine (MJ kg^{-1}).

2.3.2 Output Energy

The total output energy was calculated from the yield of the onion per unit area using the following equation.

$$TO_E = Y \times EE_O \quad ...(10)$$

where, TO_E is total output energy (MJ ha^{-1}), Y is yield (kg ha^{-1}) and EE_O is energy equivalent of onion bulb (MJ kg^{-1}).

2.3.3 Energy Analysis

Different energy parameters such as total input energy (sources), total input energy (farm operations), energy ratio, energy productivity, and net energy for onion crop production were estimated for analysis of input-output energy relationship (Singh *et al*., 2019).

$$TI_{ES} = M_E + W_E + T_E + D_E + E_E + S_E + F_E + P_E + MA_E \quad ...(11)$$

where, TI_{ES} is total input energy for different sources (MJ ha^{-1}).

As each farm operation comprised respective direct and indirect energy sources (Table 3), summation of energy consumed by all farm operations (Eqn.12) is also equal to total input energy (sources).

$$TI_{EO} = LP_E + SO_E + I_E + MF_E + PP_E + H_E + DT_E + G_E \quad ...(12)$$

where, TI_{EO} is total input energy for operations (MJ ha^{-1}), LP_E is energy required for land preparation (MJ ha^{-1}), SO_E is energy required for sowing (MJ ha^{-1}), I_E is energy required for irrigation (MJ ha^{-1}), MF_E is energy required for manure and fertilizer application (MJ ha^{-1}), PP_E is energy required for plant protection operations (MJ ha^{-1}), H_E is energy required for harvesting (MJ ha^{-1}), DT_E is energy required for detopping (MJ ha^{-1}) and G_E is energy required for grading of onions (MJ ha^{-1}).

As TI_{ES} is equal to TI_{EO}, the total input energy is denoted as TI_E in the following sections.Ortiz-Canavate and Hernanz (1999) stated that the total energy for production units is established by addition of the partial energies of each input reffered to the unit of production. After the yield produced is obtained, it is possible to calculate the energy ratio (E_R), energy productivity (E_P), net energy gain or net energy production (N_E) as described below.

Energy ratio (E_R) is defined as the ratio of caloric heat of the output products and the total sequestered energy in the production factors (Ortiz-Canavate & Hernanz, 1999). This index indicates the influence of inputs energy in obtaining output energy. The energy ratio in a process can be improved either through reduction of the energy sequestered in the inputs by optimization of energy use or through increase in the yield of product by reducing the losses.

$$E_R = \frac{TO_E}{TI_E} \quad ...(13)$$

Energy productivity (E_P) is the measure of the amount

Table 3. Source-wise and operation-wise energy input of onion cultivation in Northern Dry Zone of Karnataka

Unit Operations	Direct Energy (MJ ha^{-1})					Indirect Energy (MJ ha^{-1})				Total (MJ ha^{-1})
	Men	Women	Tractor	Diesel	Electricity	Seed	Fertilizer	Pesticide	Machinery	
Land Preparation	30.11	0.00	185.01	2685.99	0.00	0.00	0.00	0.00	3.88	2904.98
Sowing	9.76	0.00	33.58	306.15	0.00	119.3	0.00	0.00	1.73	470.52
Manure & Fertilizer	82.71	36.44	0.00	0.00	0.00	0.00	8849.26	0.00	0.00	8968.41
Plant Protection	61.98	1374.58	0.00	0.00	16.36	0.00	0.00	88.68	0.34	1541.93
Irrigation	134.12	0.00	0.00	0.00	3967.08	0.00	0.00	0.00	34.70	4135.90
Harvesting	25.49	563.75	0.00	0.00	0.00	0.00	0.00	0.00	0.00	589.23
Detopping	62.66	551.74	0.00	0.00	0.00	0.00	0.00	0.00	0.00	614.40
Grading	116.65	440.27	0.00	0.00	0.00	0.00	0.00	0.00	0.00	556.92
Total (MJ ha^{-1})	523.46	2966.78	218.58	2992.14	3983.44	119.3	8849.26	88.68	40.65	19782.29

of a product obtained per unit of input energy (Ortiz-Canavate & Hernanz, 1999). EP depends on agricultural product, location, and time. It helps to evaluate how efficiently energy is utilized in different production systems that yield a particular product (Ortiz-Canavate & Hernanz, 1999).

$$E_P = \frac{Y}{TI_E} \quad ...(14)$$

Net energy gain or net energy production (N_E) is the difference between the gross energy output produced and the total input energy required for obtaining it. This energy is normally related to the unit of production in agricultural processes (Ortiz-Canavate & Hernanz, 1999).

$$N_E = TO_E - TI_E \quad ...(15)$$

3. RESULTS AND DISCUSSION

Results showed that, the total input energy required for onion cultivation was 19,782.29 MJ ha^{-1} (Table 3). In terms of operation wise energy utilization, the highest energy consuming operations were fertilizer application (8,968.41 MJ ha^{-1}) followed by irrigation (4,135.90 MJ ha^{-1}), land preparation (2,904.98 MJ ha^{-1}), plant protection (1,541.93 MJ ha^{-1}), detopping (614.40 MJ ha^{-1}), harvesting (589.23 MJ ha^{-1}), grading (556.92 MJ ha^{-1} and sowing (470.52 MJ ha^{-1}). The highest energy input sources were observed to be fertilizer (8,849.26 MJ ha^{-1}) followed by electricity (3,983.44 MJ ha^{-1}), diesel (2,992.14 MJ ha^{-1}), and women (2,966.78 MJ ha^{-1}) (Table 3).

It was observed that for fertilizer application (45.58%), energy from fertilizer (44.98%) was the major source of energy input whereas other constituent energies *viz*., men, women and others were 0.42%, 0.19%, and 0%, respectively. Ozbek *et al*. (2021) and Allali *et al*. (2017) reported similar results of highest fertilizer consumptions of 60.43%, and 20.24% among all the indirect energy sources in their studies conducted in Adiyaman Province of Turkey and in El Hajeb Province of Morocco on onion cultivation, respectively.

As the fertilizer energy was observed to be the highest consumed source in present study, the data collected from famers was compared with the Package of Practice for onion cultivation recommended by University of Agricultural Sciences, Bagalkote, Karnataka located in this region (Anonymous, 2022) and is presented in Table 4. From Table 4, it is evident that farmers are applying 1.5 to 2 folds over the recommended fertilizer dose, which resulted less energy ratio as energy ratio is the ratio between total output energy and total input energy.

Similarly, for irrigation (21.02%), energy from electricity (20.16%) was the major source and other constituent energies *viz*., men, machinery and others were 0.68%, 0.18%, 0%, respectively. As electricity serves as the primary power source for irrigation pumps in the study area, irrigation resulted in highest consumption of electrical energy Allali *et al*. (2017) also reported similar results with highest energy consumption for irrigation 69.66% (butane-53.34%, electricity 6.44% and irrigation water 9.88%) in their study conducted in El Hajeb Province of Morocco on onion cultivation.

In case of land preparation (14.76%), energy from diesel (13.65%) was the major source whereas other constituent energies *viz*., men, tractor, machinery and others were 0.15%, 0.94, 0.02%, and 0%, respectively. Similar results were reported by Ozbek *et al*. (2021) conducted in Adiyaman Province of Turkey and by Allali *et al*. (2017) in El Hajeb Province of Morocco on onion cultivation. High diesel energy consumption is due to 100% mechanization of land preparation using tractor operated agricultural implements (Single

Table 4. Comparison of fertilizer dosage and yield between PoP of UHS, Bagalkote and farmers' practice

Input/output parameter	As per farmers' practice recorded during survey	As per Package of Practice of University of Horticultural Sciences, Bagalkote, Karnataka	Difference in usage
Nitrogen, kg ha^{-1}	250	125	Two folds higher
Phosphorous, kg ha^{-1}	125	75	1.5 folds higher
Potassium, kg ha^{-1}	125	125	-
Yield, tonne ha^{-1}	13	25	Two folds lesser

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plough, Five tyne cultivator, Nine tyne cultivator and Rotavator) in this study area. The next high energy utilising operation was plant protection (7.84%), constituting 6.99% of women, 0.45% of pesticide, 0.32% of men, 0.08% of electricity, and 0% of other energy sources.

From Fig. 2, it was further observed that significant quantity (15.08%) of women energy was consumed in onion cultivation. The major women energy-based operations were weeding (6.99%), harvesting 2.87%), detopping (2.80%) and grading (2.23%). These results suggest the need for development of machine for weeding in onion cultivation to reduce women energy as well as total input energy. Further, onion growing farmers need to be educated about the availability of onion harvesting machine, onion detopping machine and onion grading machine for adoption.

It is also observed that energy utilised by machinery was the lowest among all the input energy sources as 0.21%. The results obtained were in accordance with the results reported by Lohan *et al*. (2023) in their study conducted in Punjab State, India. They have reported that the energy consumed by machinery as 0.39%, 0.51%, 0.59%, 0.57% and 0.57% for cultivation of potato, tomato, muskmelon, garden pea and cauliflower crops, respectively. Though there are nine farm operations (Table 1) in onion cultivation, irrigation pumps had the major share of 0.18% followed by land preparation implements (0.02% and bund former (0.01%). Though the energy consumed for land preparation was 0.02%, this farm operation is completely mechanized by tractor operated tillage implements. However, this resulted in highest diesel energy consumption (third in order). Thus, this study revealed that tractors run by alternate fuels need to be developed which would further enhance the energy ratio in onion cultivation and environment friendly crop production system.

3.1 Estimation of Energy Parameters

The energy parameters *viz*., direct, indirect, total input, output energy, energy ratio, energy productivity and net energy calculated for onion cultivation in major onion growing area of Karnataka state is presented in Table 5. The total output energy for onion cultivation was 21,826.40 MJ ha^{-1} as against the total input energy of 19,782.29 MJ ha^{-1}. The utilization of direct input energy (54.10%) was higher than indirect input energy (45.90%). Energy ratio, energy productivity and net energy production for onion cultivation were

Table 5. Energy parameters for onion production in major onion growing region of Karnataka state

Energy parameters	Value
Direct energy, MJ ha^{-1}	10684.40 ±722.38
Indirect energy, MJ ha^{-1}	9097.89 ± 465.90
Total input energy, MJ ha^{-1}	19782.29 ± 914.59
Yield, kg ha^{-1}	13641.50 ± 1146.80
Total output Energy, MJ ha^{-1}	21826.40 ± 1834.87
Energy ratio	1.11 ± 0.11
Energy productivity, kg MJ^{-1}	0.69 ± 0.07
Net energy, MJ ha^{-1}	2044.11 ± 2001.87

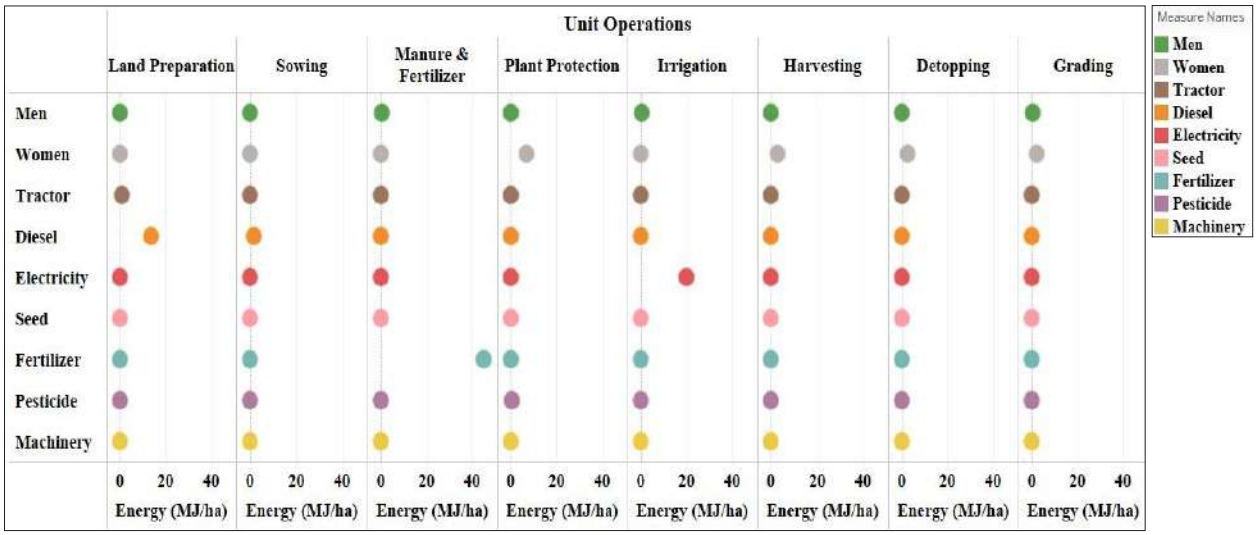

Fig. 2. Operation-wise and source-wise energy utilised in onion cultivation

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calculated as 1.11, 0.69 kg MJ^{-1} and 2,044.11 MJ ha^{-1}, respectively. As the energy ratio (1.11) is more than 1.0, the cultivation of onion is profitable in terms of energy consumption. Previous studies reported energy use efficiency (EUE) for onion cultivation as 2.21 in Adiyaman Province of Turkey (Ozbek *et al*., 2021), and 0.78 in El Hajeb province of Morocco (Allali *et al*., 2017). Our result of energy productivity (0.69 kg MJ^{-1}) is almost similar to the findings of Hajeb province of Morocco (0.54 kg MJ^{-1}) but the energy productivity value is very low as compared to the value reported for Adiyaman Province of Turkey, where energy productivity is 1.38 kg MJ^{-1}. This variation in our results and that of Adiyaman Province of Turkey (Ozbek *et al*., 2021) is due to the fact that though the input energy of onion cultivation in the present study (19,782.29 MJ ha^{-1}) is comparable (22,463.52 MJ ha^{-1}) with that of the Tekirdag province of Turkey, but, the output energy [49,704 MJ ha^{-1} (31.065 t ha^{-1})] in case of the Tekirdag province of Turkey was comparatively higher as compared to the present study [21,826.40 MJ ha^{-1} (13.641 t ha^{-1})]. The low output energy in the present study is due lower onion yield (13641.50 kg ha^{-1}) as compared to the yield (31,065 kg ha^{-1}) obtained in the Tekirdag province of Turkey. Considering national productivity (16.34 t ha^{-1}), the energy ratio of the present study area could be achieved up to 1.32. The reason for low productivity in this region might be due to varied soil fertility, climatic condition and cultivation practices followed by the farmers. In order to achieve higher EUE in onion cultivation, the onion productivity needs to be increased through high yielding onion varieties, improving soil health, newer production technologies and mechanization.

4. CONCLUSIONS

This study was conducted to assess the energy utilization for onion cultivation in the Northern Dry Zone of Karnataka, which is the highest onion growing region in Karnataka state. The total input energy for cultivation of onion in this study was 19,782.29 ± 914.59 MJ. ha^{-1} and output energy was 21,826.40 ± 1834.87 with an energy ratio of 1.11 ± 0.11. Significant energy utilizing sources were fertilizer of 8,849.26 MJ ha-^{1} (44.98%), electricity of 3,983.44 MJ ha^{-1} (20.25%), diesel of 2,992.14 MJ ha^{-1} (15.21%) and women of 2,966.78 MJ ha^{-1} (15.08%). From this study it was also evident that farmers apply higher than the recommended fertilizer dose resulting less energy ratio. Hence, farmers should be educated for optimal fertilizer application which would also further ensure soil health. The highest share of electricity use was for operating irrigation pumps. Use of solar powered irrigation pumps would result in reduced electricity and diesel energy use for irrigation. Research on alternative fuels for operating tractors needs to be strengthened. It reduces use and dependency of diesel energy and further help in cleaner onion crop cultivation. The study also revealed that contribution of women energy in inter cultivation operation was significant. From this, it is evident that development of weeders for onion cultivation will reduce the women input energy. Awareness generation of the recently developed machinery *viz*., onion harvester, onion detopping machine and onion grader developed by research institutes is a need for wider adoption. By addressing these findings, the energy ratio in onion cultivation can be enhanced.

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AUTHOR CONTRIBUTIONS

A. Carolin Rathinakumari: Conceptualization; methodology; data curation; draft writing; formal analysis, project administration; **G. Senthil Kumaran:** Concept**;** data curation; project administration; **S. A. Venu:** Data collection; draft manuscript preparation; **B. S. Ashok Suraj:** Data collection and arrangement of data; **Murugesh Huchagoudar:** Data analysis & data curation; **A. R. Surendra:** Graphical representations and manuscript preparation**.**

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest in any form that could have influences the research work reported in this paper. Competing interest in disclosure of the research work has been considered by the authors.

DATA AVAILABILITY STATEMENT

Data will be made available by the corresponding author on receiving a reasonable request.

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